A PROJECT REPORT

on

"AIR QUALITY INDEX PREDICTION USING MACHINE LEARNING ALGORITHMS"

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In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN INFORMATION TECHNOLOGY

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CERTIFICATE

This is certify that the project entitled

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is a record of bonafide work carried out by them, in the partial fulfillment of the requirement for the award of Degree of Bachelor of Engineering (Computer Sci-ence & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2022-2023, under our guidance.

Dr. Minakhi Rout
Project Guide

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Introduction

Pollution is a word derived from ‘polluere’ which in latin means contamination in general sense of something that polluted the surroundings and the environment is called pollution. The presence of harmful substances in the water,soil,air which are affecting the lives of the common people and the environment is also defined as pollution.These harmful substances in turn which affect the environment are commonly known as pollutants or defined as one. Human activities and development are one major reason for affecting the biological diversity of the ecosystem and in turn cause pollution also posing a threat to the Environmental sustainability.

Air Pollution: - It can be defined as the contamination of the natural air which gets mixed with different pollutants which may include harmful fumes and chemicals. This type of contamination can be caused by burning material or by gasses emitted by vehicles or harmful fumes emitted as a by-product of industries. The biggest side effect or drawback of air pollution is Global warming according to the experts.

Types of Pollutants

Primary Pollutants:

Primary pollutants are emissions directly into the air or environment from an identifiable source.From places such as vehicles, industrial processes, construction and natural events such as fire and explosion.

Examples include particulate matter (PM), carbon monoxide (CO), sulfur dioxide (SO2), nitrogen oxides (NOx) and volatile organic compounds (VOC).

This substance, generally defined by its form, can directly affect the environment and human health when released. It especially causes air pollution, smoke formation, respiratory diseases and environmental degradation. Often effects occur that affect the immediate area of the emission point.

Secondary pollutants:

Secondary pollutants are not released directly. Instead, they form as a result of chemical reactions in the atmosphere involving major pollutants, sunlight, and other atmospheric factors. It is formed from chemical compounds that include major pollutants such as nitrogen oxides, harmful organic compounds and sunlightExamples include ground-level ozone (O3), secondary aerosols, and some types of acid rain.

They are difficult to control because they form depending on various weather conditions and interactions. Initially affects smog formation, including ground-level ozone and the formation of secondary aerosols. Major impact on human health, ecosystems and resources. may spread for a long time from where it was created and cause regional and international problems.

To measure the AQI the following pollutants are taken into account

Particulate count (PM2.five and PM10):
Definition: PM2.5 refers to first-rate inhalable debris with a diameter of 2.5 micrometers or smaller, whilst PM10 includes slightly larger particles with a diameter of 10 micrometers or smaller.
traits: these particles come from diverse sources, along with automobile emissions, commercial strategies, construction, and natural assets like dirt storms and wildfires.
effect: Inhalation of PM2.5 and PM10 can cause respiratory and cardiovascular problems, with PM2.five having a higher capacity to penetrate deep into the lungs.

Ozone (O3):
Definition: ground-level ozone (O3) is a secondary pollutant formed by way of chemical reactions among risky natural compounds (VOCs) and nitrogen oxides (NOx) within the presence of sunlight.
traits: It bureaucracy because of complex atmospheric reactions, in particular in urban regions with high emissions of VOCs and NOx.
effect: breathing ozone can cause breathing problems, aggravate allergies, and decrease lung features.

Carbon Monoxide (CO):
Definition: Carbon monoxide is a drab, odorless fuel produced via incomplete combustion of fossil fuels and biomass.
characteristics: in particular emitted by way of vehicles, commercial approaches, and home heating structures.
impact: CO can intervene with the frame's capability to move oxygen, leading to headaches, dizziness, and, at high levels, may be existence-threatening.

Sulfur Dioxide (SO2):
Definition: Sulfur dioxide is a gasoline produced through burning fossil fuels containing sulfur compounds, particularly coal and oil.
traits: Emitted through strength flora, business facilities, and volcanic eruptions.
impact: Can cause respiratory problems and make contributions to acid rain, that may harm ecosystems and infrastructure.

Nitrogen Dioxide (NO2):
Definition: Nitrogen dioxide is a reddish-brown fuel produced via excessive-temperature combustion, particularly in vehicles and strength flowers.
traits: usually emitted in urban regions with heavy visitors and industrial activities.
impact: NO2 can worsen the respiratory machine, worsen allergies, and make a contribution to the formation of smog.

Causes of Pollution

Industrial sports:
Heavy industries along with manufacturing, mining, and refining launch various pollutants into the air, water, and soil. Emissions from factories and production processes contribute extensively to environmental pollutants.

Automobile Emissions:
The combustion of fossil fuels in vehicles, vans, and other transportation modes releases pollution like carbon monoxide, nitrogen oxides, and particulates. high traffic regions regularly exhibit accelerated air pollution levels.

Agricultural Practices:
Agricultural sports, which include using fertilizers, insecticides, and animal waste, cause soil and water infection. Runoff from farms contains pollutants into our bodies, causing water pollutants.

Mistaken Waste Disposal:
incorrect disposal of stable waste, which includes plastics, electronic waste, and household garbage, contaminated land and water. Landfills and incineration contribute to soil, air, and water pollutants if not controlled nicely.

Deforestation and Land Use modifications:
Clearing forests for agriculture, urbanization, and infrastructure development disrupts ecosystems, mainly to soil erosion and loss of biodiversity. Such changes contribute to air and water pollutants. Extractive sports like mining and quarrying launch hazardous substances and pollution into the environment, affecting soil, water quality, and air purity.

Creation and Demolition:
production sports produce dirt, debris, and chemical emissions that impact air first-class. Demolition methods launch pollutants and construction waste, contributing to environmental degradation.

Natural occasions:
Natural failures like wildfires, volcanic eruptions, and dust storms can release full-size pollutants into the ecosystem, affecting air excellent regionally or even globally.

Energy production:
The burning of fossil fuels for electricity technology in power vegetation and heating contributes substantially to air pollution via the emission of greenhouse gases and different pollutants.

Need of AQI

The Air quality Index (AQI) serves as a critical tool for gauging and communicating the excellence of the air we breathe. right here's a completely unique tackle its necessity:

Public fitness recognition:
AQI gives a simplified and comprehensible degree of air quality, allowing the general public to grasp the capacity health risks associated with distinct pollution stages. It facilitates people, in particular susceptible populations, to make knowledgeable choices to defend their fitness.

Guidance for protective Measures:
By way of categorizing air nice into special ranges (true, mild, dangerous, and many others.), the AQI gives steerage on vital precautions. It advises human beings on adjusting outdoor sports, sporting masks, or staying indoors for the duration of excessive pollutant periods to limit publicity to harmful pollution.

Environmental monitoring and law:
It aids environmental businesses and policymakers in tracking pollutants tiers throughout areas, permitting them to enforce focused interventions and regulations. This facts-pushed approach allows in placing emission standards and formulating regulations to enhance air first-class.

Impact on Economies and making plans:
AQI records affect urban planning, infrastructure development, and financial choices. cities with better air high-quality may attract companies and citizens, whilst people with poor air quality may additionally face demanding situations in phrases of public health expenses and reduced splendor.

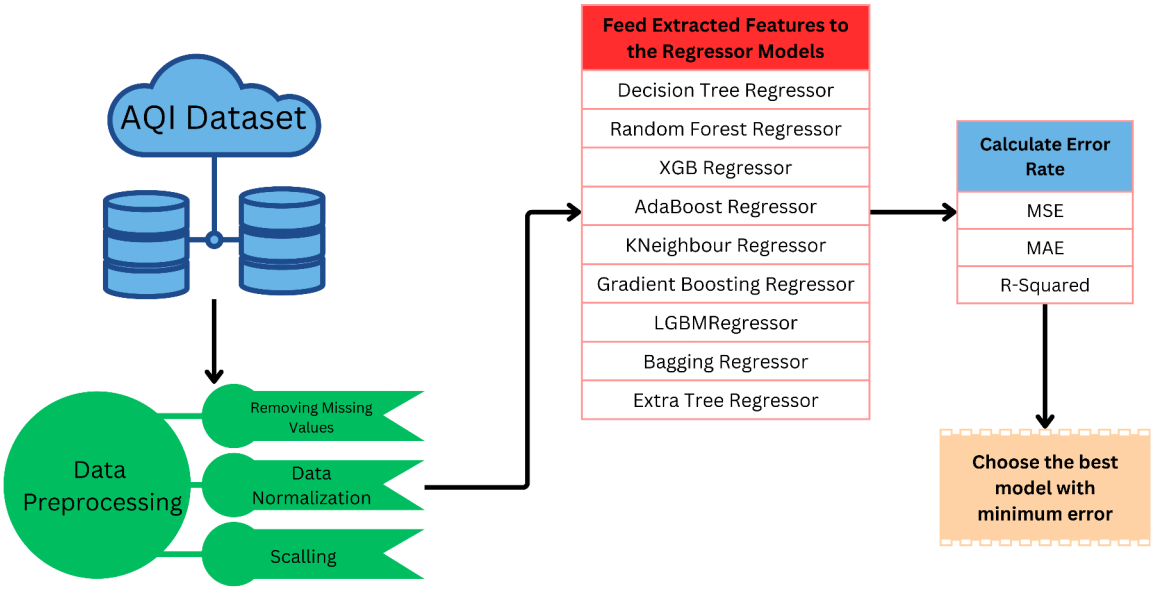
Emergency response and Preparedness:
All through pollution peaks or environmental emergencies, the AQI serves as a critical tool for emergency response making plans. It triggers well timed alerts and measures to shield public fitness, specially in situations like wildfires or business injuries.

Worldwide Cooperation and awareness:
The AQI standardizes air satisfactory assessment globally, facilitating international cooperation in addressing transboundary pollution troubles. It promotes consciousness about the interconnectedness of air quality and environmental fitness on an international scale.

The AQI's role extends past mere size; it empowers individuals, guides rules, and fosters collaboration towards a common goal of healthier environments and stepped forward public properly-being.

2.Methodology/ Proposal

2.1 Design model



2.2 Implementation Platform

This test was conducted on an online compiler called Kaggle, using an Intel Core i7-8550U CPU @ 1.99 GHz, with an AMD Radeon 530 graphics card and 13 GB of memory. It offers a unique blend of community collaboration, real-world challenges, and resources.

2.3 Libraries Used

The NumPy library is used for numeric operations and arrays, while pandas is used for data manipulation and analysis. The matplotlib library is used for data visualization and plotting. The os library provides a way to interact with the operating system, the sci-kit-learn library is used for evaluating model performance. The Seaborn library creates visualizations, specifically heatmaps, for confusion matrices.

2.4 Data Set Description

The dataset comprises air quality information, including parameters such as PM2.5, PM10, O3, SO2, CO, wind speed, humidity, temperature, AQI (Air Quality Index), and AQI Type that has been scraped from a website called ‘Air matters’ using the python beautiful-soup library. Some entries have missing values, prompting the need for normalization techniques.

2.4.1 AQI Range

The Air Quality Index (AQI) is a measure used to communicate how polluted the air currently is or how polluted it is forecast to become. Different countries may have different AQI scales and corresponding health advisories. In India, the National Air Quality Index (NAQI) is used to assess and communicate air quality to the public. The NAQI categorizes air quality into different bands, each associated with a specific health advisory. As of my last knowledge update in January 2022, the NAQI has six categories, ranging from "Good" to "Severe

AQI Range	Color code	Possible Health Impacts
0-50	Good	Minimal impact
51-100	Satisfactory	Minor breathing discomfort to sensitive people
101-200	Moderate	Breathing discomfort to the people with lungs, asthma and heart diseases
201-300	Poor	Breathing discomfort to most people on prolonged exposure
301-400	Very Poor	Respiratory illness on prolonged exposure
401-500	Severe	Affects healthy people and seriously impacts those with existing diseases

2.4.2 Normalization

Some entries in our dataset have missing values, prompting the need for normalization techniques.

1. **Iterative Imputer**
   The Iterative Imputer process is a technique for filling in missing values in a dataset by making multiple passes over the data and iteratively estimating the missing values based on observed values and relationships between variables. This process uses machine learning models to make educated guesses about missing numbers until a certain level of agreement is reached. The scikit-learn library in Python provides an implementation of Iterative Imputer. The Iterative Imputer process is particularly useful when dealing with datasets where missing values are not randomly distributed and have some underlying patterns or relationships with other variables. The imputer iteratively refines its

estimates based on the observed values in the dataset, making it a powerful tool for handling missing data in a more sophisticated manner compared to simple imputation techniques.

2. **KNN Imputer**
   The KNN (K-Nearest Neighbors) Imputer is a method for filling in missing values in a dataset by estimating them based on the known values of their nearest neighbors. In the context of this implementation, the scikit-learn library in Python provides a KNNImputer class that can be used for this purpose.This Imputer is advantageous when the distribution of missing values is not random and when there are patterns or correlations in the data that can be leveraged to make more accurate imputations. It uses the information from neighboring data points to estimate missing values, providing a more sophisticated approach compared to simple imputation techniques like mean, median, or mode.
   It was determined that KNN-Imputer worked best for normalizing our data.

2.4.3 Scaling

Different Scaling techniques are employed and are evaluated based on their impact on model performance.
The dataset contains features such as PM2.5, PM10, O3, SO2, CO, Wind Speed, and Temperature, along with the target variable AQI. The State and City columns have been dropped as they are considered non-contributory.
Different feature scaling methods are applied to the input features (X) to standardize their scales and improve the performance of the AQI prediction model.

1. **Min-Max Scaling**
   Min-Max Scaling, also known as normalization, transforms the values of each feature to a specific range, usually between 0 and 1. It is calculated by subtracting the minimum value of the feature ($X_{\min}$) and then dividing by the range of the feature ($X_{\max} - X_{\min}$). This ensures that the transformed values lie within a standardized scale.

   Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	0.099099	0.101266	0.0625	0.036364	0.090258	0.166667	0.70
1	0.315315	0.310127	0.3250	0.236364	0.227193	0.583333	0.70
2	0.306306	0.310127	0.2875	0.127273	0.144102	0.694444	0.70
3	0.315315	0.322785	0.2625	0.163636	0.151656	0.500000	0.70
4	0.324324	0.34772	0.2750	0.127273	0.142940	0.611111	0.75

2. **Standardization**

Standardization scales the values of each feature to have a mean of 0 and a standard deviation of 1. It is achieved by subtracting the mean of the feature and then dividing by the standard deviation. This process centers the data around zero and ensures that it has a consistent scale.

Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	-1.126445	-1.268933	-0.894682	-0.916490	-0.807918	-0.652965	0.462246
1	0.438475	0.183351	0.541981	0.291239	-0.144186	0.936501	0.462246
2	0.373270	0.183351	0.336744	-0.367522	-0.546931	1.360358	0.462246
3	0.438475	0.271368	0.199919	-0.147935	-0.510318	0.618608	0.462246
4	0.503680	0.403394	0.268331	-0.367522	-0.552564	1.042465	0.780138

3. **Robust Scaling**

Robust Scaling is similar to standardization but is more robust to outliers. It uses the interquartile range (IQR) instead of the standard deviation. The values are scaled by subtracting the first quartile ($Q1$) and then dividing by the interquartile range ($Q3-Q1$). This method is less sensitive to extreme values in the dataset.

Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	-0.8750	-1.00000	-0.722222	-1.000000	-0.480899	-0.3125	0.25
1	0.6250	0.222222	0.444444	0.571429	0.048689	0.6250	0.25
2	0.5625	0.222222	0.277778	-0.285714	-0.272659	0.8750	0.25
3	0.6250	0.296296	0.166667	0.000000	-0.243446	0.4375	0.25
4	0.6875	0.407407	0.222222	-0.285714	-0.277154	0.6875	0.25

4. **Log Transformation**

Log Transformation is applied to features that exhibit a skewed distribution. It helps stabilize the variance and normalize the data by taking the natural logarithm of each value plus one. This is particularly useful when the data spans several orders of magnitude.

Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	2.564949	3.044522	1.945910	1.386294	5.376820	2.079442	3.367296
1	3.610918	3.988984	3.332205	2.708050	6.113682	3.135494	3.367296
2	3.583519	3.988984	3.218876	2.197225	5.733341	3.295837	3.367296
3	3.610918	4.025352	3.135494	2.397895	5.774552	2.995732	3.367296
4	3.637586	4.077537	3.178054	2.197225	5.726848	3.178054	3.401197

5. **Power Transformation**
   The Power Transformer, using the Yeo-Johnson method, is a family of power transformations that aim to stabilize variance and make the data more Gaussian-like. It handles both positive and negative values and automatically determines the appropriate transformation based on the data distribution.

   Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	-1.209503	-1.436565	-0.889310	-1.362830	-0.805709	-0.471743	0.402798
1	0.532714	0.304464	0.699772	0.607013	0.140319	0.971267	0.402798
2	0.472946	0.304464	0.536323	-0.175296	-0.363215	1.230305	0.402798
3	0.532714	0.387636	0.420466	0.128793	-0.310263	0.754812	0.402798
4	0.591770	0.509415	0.479144	-0.175296	-0.371524	1.038888	0.777663

6. **Decimal Scaling**
   Decimal Scaling involves dividing each value by an appropriate power of 10. This ensures that the values fall within a specific range, such as [-1, 1] or [0, 1], without losing the relative differences between the values.

   Results after Scaling:-

	PM2.5	PM10	O3	SO2	CO	WIND SPEED	TEMP
0	0.12	0.20	0.06	0.03	2.15	0.07	0.28
1	0.36	0.53	0.27	0.14	4.51	0.22	0.28
2	0.35	0.53	0.24	0.08	3.08	0.26	0.28
3	0.36	0.55	0.22	0.10	3.21	0.19	0.28
4	0.37	0.58	0.23	0.08	3.06	0.23	0.29

Following a thorough comparison of Mean Squared Error (MSE) scores for each technique, it was discerned that **Power Transformer** emerged as the most effective choice for our dataset.

2.4.3 Train Test Split

The dataset is split into training and testing sets using the `train_test_split` function from scikit-learn. The split ratio is set to 80% training and 20% testing, with a random seed of 42 for reproducibility. The chosen feature scaling method, determined earlier as "Power Transformer," is applied to both the training and testing sets. If scaling is required (not None), the scaler is fitted on the training data, and the transformation is applied to both the training and testing sets. If no scaling is necessary, the original sets are retained.

The successful implementation of the data split and scaling process lays the foundation for subsequent model training and evaluation. The splitting ensures that the model is trained on a diverse set of data and tested on a separate, unseen dataset. The scaling process, guided by the previously identified optimal method ("Power Transformer"), ensures that the features are on a comparable scale, aiding the model in making accurate predictions. With the dataset split and scaled, the next steps involve model training and evaluation. The preprocessed data sets (`X_train_scaled`, `X_test_scaled`, `y_train`, `y_test`) are ready for feeding into the chosen machine learning algorithm. The model's performance will be assessed using appropriate metrics, such as Mean Squared Error or R-squared, to determine its effectiveness in predicting the Air Quality Index.

2.6 Models Included

A continuous target variable can be predicted using regression analysis, a statistical method, based on one or more independent variables. Regression models come in a multitude of forms, each with unique advantages and disadvantages. The Decision Tree Regressor, Random Forest Regressor, XGBoost Regressor, AdaBoost Regressor, KNeighbors Regressor, Gradient Boosting Regressor, LightGBM Regressor, Bagging Regressor, and ExtraTrees Regressor are the nine regression models that will be compared in this report.

Regressor Decision Tree

A decision tree is a structure with nodes and branches that resembles a tree. Every branch indicates a potential result of a decision, and every node represents a decision. Recursively dividing the data into smaller subsets according to the values of the independent variables is how a decision tree predicts the target variable.

The Random Forest Regressor

An ensemble approach that combines several decision trees is called a random forest. A slightly different subset of the data is used to train each decision tree in the random forest, and the average of all the trees' forecasts is used to get the final prediction.

XGBoost Regressor

Another ensemble technique that combines boosting with decision trees is called XGBoost. Boosting is a method that trains decision trees in a sequential fashion, with each tree learning from the mistakes made by the trees that came before it.

AdaBoost Regressor

Similar to XGBoost, AdaBoost is an ensemble approach that employs a slightly different boosting algorithm.

Regressor K Neighbors

A nonparametric regression model called a KNeighbors regressor makes predictions about a new data point's value by analyzing the values of its k nearest neighbors in the training set..

LGBM Regressor

Similar to Gradient Boosting Regressor and XGBoost, another ensemble technique is LGBM. It is renowned for being quick and effective.

Baggage Regressor

An ensemble technique that combines several copies of the same regression model is called a bagging regressor. The model is trained on distinct bootstrap samples of the data for each copy, and the average of all the copies' predictions is used to get the final prediction etc

2.7 Hyperparameter Used

The parameter grids for each model are specified in a dictionary. The keys of the dictionary are the names of the parameters, and the values of the dictionary are lists of possible values for each parameter.
The models included in the array are: DecisionTreeRegressor, RandomForestRegressor, XGBRegressor , AdaBoostRegressor , KNeighborsRegressor , GradientBoostingRegressor LGBMRegressor, BaggingRegressor, ExtraTreesRegressor

DecisionTreeRegressor: A decision tree is a basic machine learning model that divides the data into ever-smaller subsets recursively in order to generate predictions. A node in the tree represents each partition, and for every node, there is a decision rule that dictates which child node it should follow. The final forecasts are represented by the leaves on the tree.

RandomForestRegressor: A collection of decision trees is called a random forest. A distinct subset of data is used to train each decision tree in the forest, and the final forecast is derived by averaging the predictions of the trees. Because they are less prone to overfitting than decision trees, random forests provide results that are more accurate.

KNeighborsRegressor: The K-nearest neighbors machine learning model is a non-parametric model that predicts data based on how similar the input data is to the training data. For problems involving regression and classification, K-nearest neighbors provides a straightforward and efficient approach.

GradientBoostingRegressor: An ensemble learning technique that combines the predictions of several decision trees is known as a gradient boosting regressor. It is well known that gradient boosting regressors are accurate and scalable.

ExtraTreesRegressor: This kind of ensemble learning technique aggregates the forecasts from several additional trees. A decision tree variant known as extra trees employs a random subset of features at each split. Speed and precision are well-known attributes of ExtraTreesRegressor. etc

3. Result Analysis

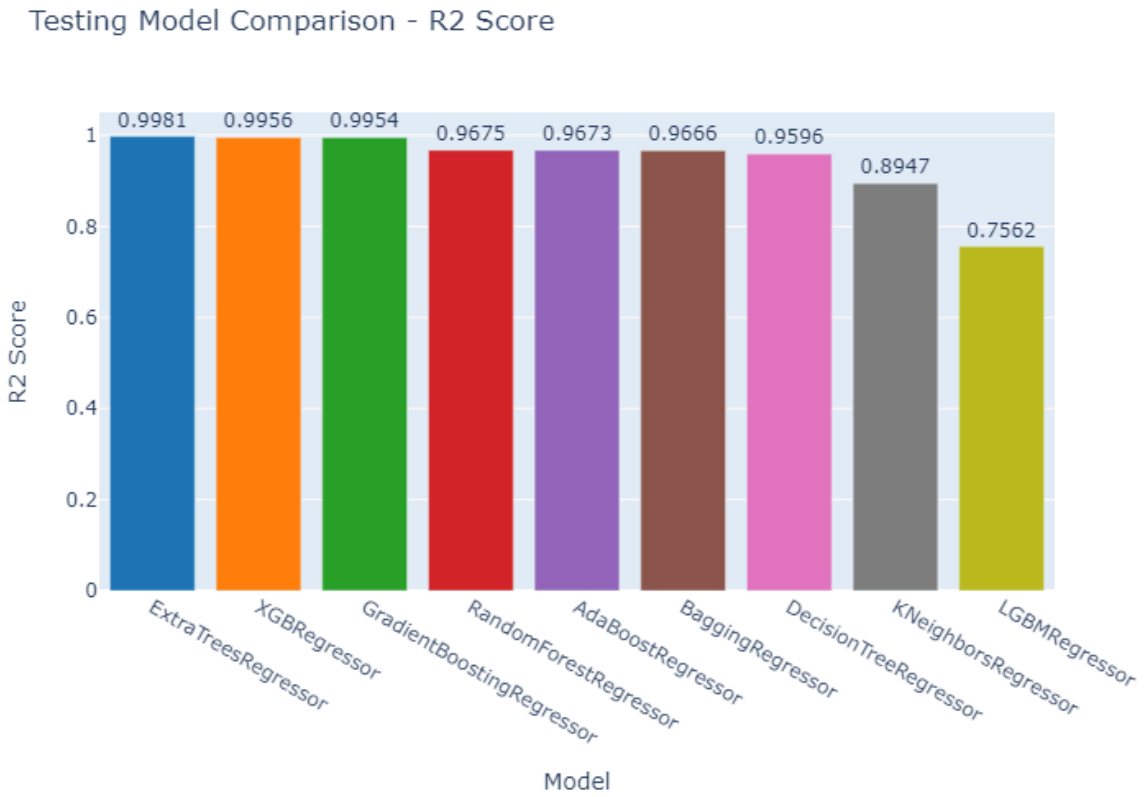
The metrics used in the proposed work are R-SQUARE,mean squared error (MSE) and mean absolute error (MAE)

3.1 R-SQUARE

R-SQUARE indicates to what extent the regression model is in line with the observed data. A higher *R* square value denotes a better model fit, the *R* Square equation is shown by equation

$$R - SQUARE = SSregr/SStt$$

The sum of squares due to regression is denoted by *SS*regr (explained sum of squares), while the sum of squares overall is denoted by *SStt*. The degree to which the regression model fits the data well is shown by the sum of squares due to regression. The total sum of squares is used to determine how much the observed data has changed (data utilized in regression modeling).



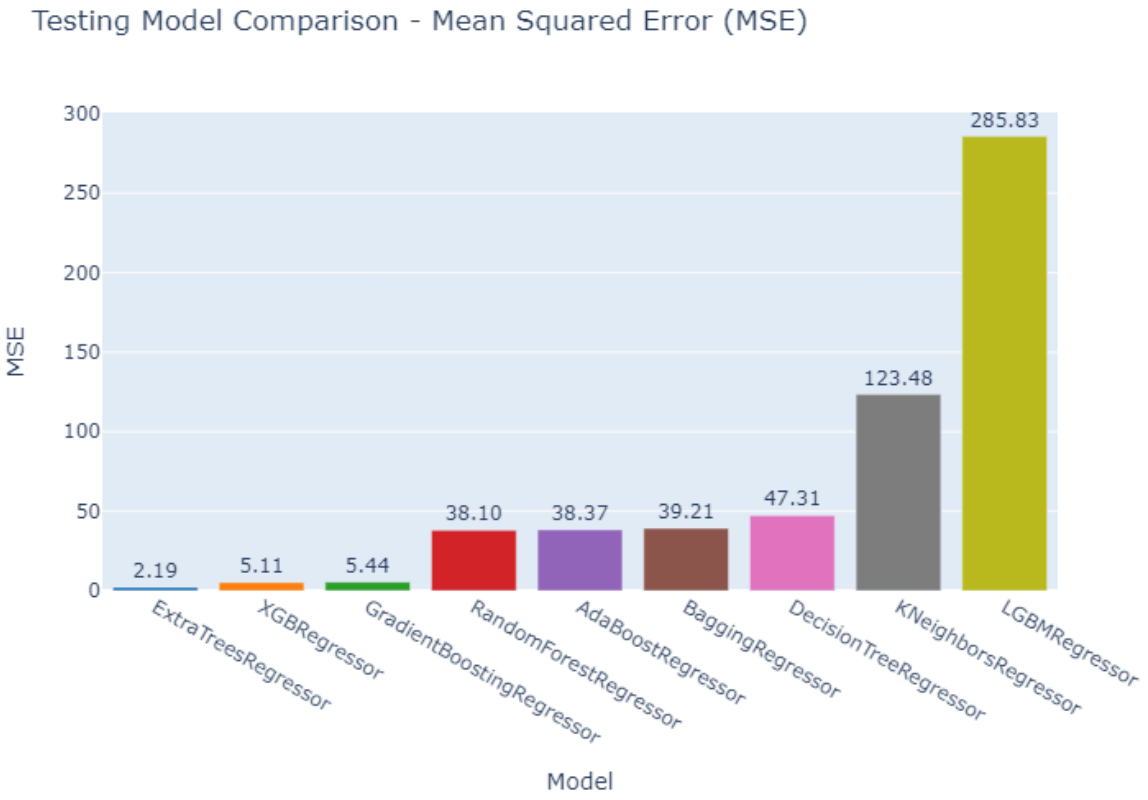
3.2 MSE

MSE is a parameter that measures how closely a fitted line resembles a set of data points. The lower the value, the closer it is to the line, and hence the better. If the MSE value = 0, the model is perfect. It is shown in equation

$$\mathbf{MSE} = \Sigma(y_i - p_i)^2/n$$

where

(a) y_i = The ith observed value
(b) p_i = The corresponding predicted value for y_i
(c) n = The number of observations



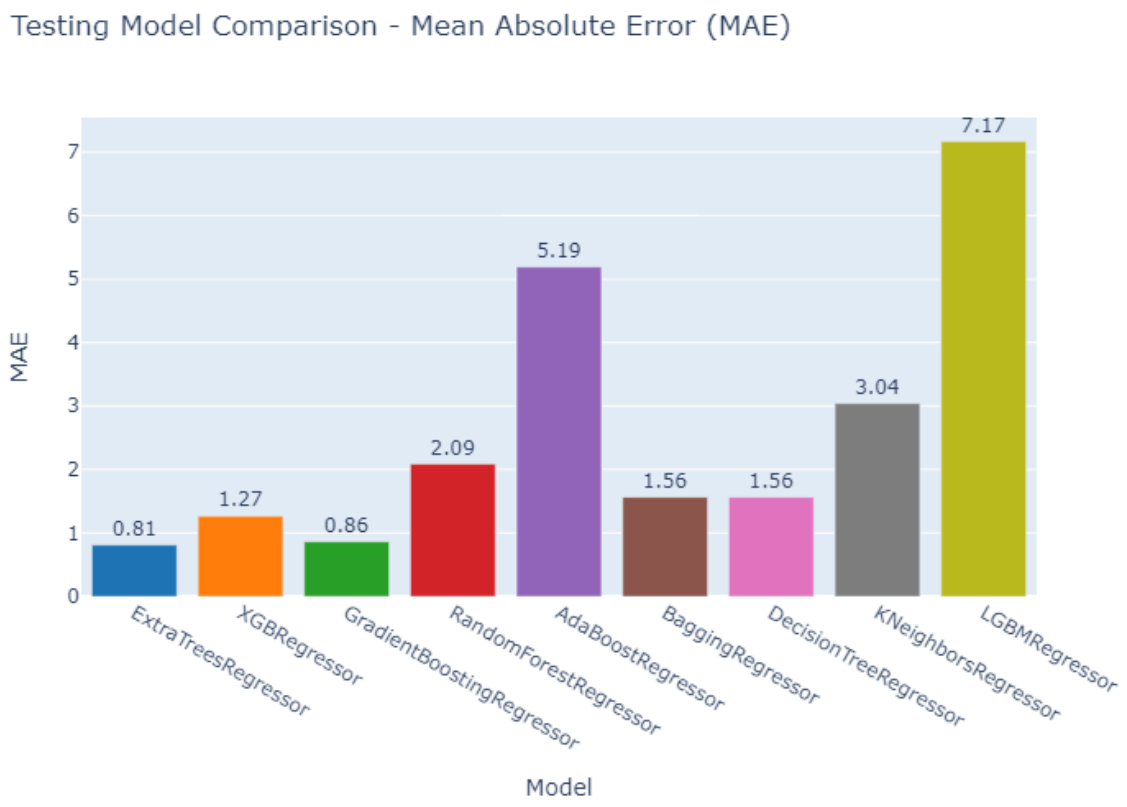
3.3 MAE

MAE evaluates the absolute distance of the observations to the predictions on the regression line. It is shown in the equation

$$\mathbf{MAE = (1/n) * \Sigma |a_i - b_i\hat{}|}$$

where

(a) a_i = The ith observed value
(b) b_i = The corresponding predicted value for y_i
(c) n = The number of observations



3.4 Analysis Comparison Between different Models

Rank	Model	MSE	MAE	R2 Score
1	ExtraTreesRegressor	2.19	0.81	99.81%
2	XGBRegressor	5.11	1.27	99.56%
3	GradientBoostingRegressor	5.44	0.86	99.54%
4	RandomForestRegressor	38.10	2.09	96.75%
5	AdaBoostRegressor	38.37	5.19	96.73%
6	BaggingRegressor	39.21	1.56	96.66%
7	DecisionTreeRegressor	47.31	1.56	95.96%
8	KNeighborsRegressor	123.48	3.04	89.47%
9	LGBMRegressor	285.83	7.17	75.62%

Explanation:

1. ExtraTreesRegressor: This model performs the best across all metrics. It has the lowest MSE, indicating its predictions are, on average, closest to the actual values. The low MAE also supports this, and the high R2 Score (99.81%) indicates a very good fit to the data.
2. XGBRegressor and GradientBoostingRegressor: These models also perform well, with relatively low MSE and high R2 Score. The XGBRegressor has a slightly lower MSE, making it slightly better in terms of mean squared error, but the difference is not significant. Both models use boosting techniques, which sequentially improve the predictions.
3. RandomForestRegressor, AdaBoostRegressor, and BaggingRegressor: These models have higher MSE and lower R2 Score compared to the top three. RandomForestRegressor and AdaBoostRegressor have similar performance, while BaggingRegressor performs slightly worse.
4. DecisionTreeRegressor: This model has a higher MSE and lower R2 Score compared to ensemble methods like RandomForestRegressor and ExtraTreesRegressor. Decision trees can be prone to overfitting and may not generalize well to new data.
5. KNeighborsRegressor: This model has a significantly higher MSE and a lower R2 Score compared to the others. It suggests that the pattern in the data is not well captured by the nearest neighbors approach.
6. LGBMRegressor: This model has the highest MSE and the lowest R2 Score among all. LightGBM is a gradient boosting framework, but in this case, it seems to struggle with the given data.

Which is the Best Model and Why?

The Best Model is the ExtraTreeRegressor the reason being:

1. Lowest MSE (Mean Squared Error): MSE measures the average squared difference between predicted values and actual values. A lower MSE indicates that the model's predictions are, on average, closer to the true values. The ExtraTreesRegressor has the lowest MSE (2.19), which means it has the smallest overall squared error among all models.
2. Lowest MAE (Mean Absolute Error): MAE measures the average absolute difference between predicted and actual values. Like MSE, a lower MAE indicates better accuracy. The ExtraTreesRegressor also has the lowest MAE (0.81), further supporting its accuracy in prediction.
3. Highest R2 Score: The R2 Score, or coefficient of determination, represents the proportion of the variance in the dependent variable that is predictable from the independent variables. A higher R2 Score indicates a better fit of the model to the data. The ExtraTreesRegressor has the highest R2 Score (99.81%), signifying an excellent fit to the data

The ExtraTreesRegressor outperforms other models in terms of accuracy (low MSE and MAE) and goodness of fit (high R2 Score). It consistently provides the most accurate and reliable predictions based on the given evaluation metrics.

4 Conclusion and Future Scope

4.1 Conclusion:

Air quality is a major factor that affects the life and well being of people and surrounding as well as it a major factor in working of the ecosystem and environment. By predicting the air quality index people can make informed decisions about their lifestyle.The development of an air quality prediction model was undertaken,leveraging various stages ranging from data acquisition to the selection of the model,encompassing from web scraping for reliability,meticulous data preprocessing and evaluation.

Challenges encountered during the project focused on finding good weather data and dealing with irregular data (such as missing values and parameters). These issues are overcome using web scraping techniques and new data prioritization, including the use of benchmarking techniques such as KNN Imputer and PowerTransformer.

Among the various models evaluated, BaggingRegressor turned out to be the best at predicting climate quality. A low least squares error (MSE) value indicates its best performance and indicates accurate weather forecast ability.

4.2 Future Scope:

In the future, the application of our air quality prediction model will have great potential in many areas and will become an important tool for health. Climate change positively affects decisions regarding outdoor activities. City planners can use this model to develop and improve urban development strategies to reduce the impact of pollution, improve infrastructure and green spaces In healthcare, the integration of these models can help predict increased access to patients during cloud negative wind, which is beneficial for the negative group. Moreover, its contribution to promoting climate change and the collaboration of experts from different fields can create new policy and research insights. Anchoring and evaluating this international model has the potential to solve climate problems in different regions, promote regional solutions and achieve environmental and social goals.

5. References

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INDIVIDUAL CONTRIBUTION REPORT:

Air Quality Index Prediction Using Machine Learning Algorithm

SWAMITA SACCHAN
2029038

Abstract: Accurate Air Quality Index (AQI) prediction is essential for public health management as global air quality concerns grow. This work uses a collection of air quality metrics, meteorological factors, and pollutant concentrations to anticipate AQI values using machine learning regression models. Air pollution is rising due to factors including automotive emissions, industrial activities, and fossil fuel burning. This study examines the pros and cons of each regression model for AQI prediction through thorough experimentation and comparison. The findings expand environmental science machine learning applications and provide policymakers and public health experts with practical insights. This work establishes proactive air quality management techniques and emphasizes the importance of machine learning in solving air pollution's complex difficulties by determining the best models.

Individual contribution and findings: My main responsibility in the project revolve around the examining the trained models to learn about the connections between AQI forecasts and input features.Using graphical representations of model predictions and uncertainty to share insights and determining the major contributing reasons to fluctuations in the AQI and any possible ramifications I worked on the hyperparameter tuning and the models used.

Individual contribution to project report preparation: My contributions to the report include an analysis of the models employed for air quality index (AQI) prediction and a comprehensive evaluation of hyperparameter tuning strategies.

Individual contribution for project presentation and demonstration:



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INDIVIDUAL CONTRIBUTION REPORT:

Air Quality Index Prediction Using Machine Learning Algorithm

ANUSHKA AMAR
2029048

Abstract: Accurate Air Quality Index (AQI) prediction is essential for public health management as global air quality concerns grow. This work uses a collection of air quality metrics, meteorological factors, and pollutant concentrations to anticipate AQI values using machine learning regression models. Air pollution is rising due to factors including automotive emissions, industrial activities, and fossil fuel burning. This study examines the pros and cons of each regression model for AQI prediction through thorough experimentation and comparison. The findings expand environmental science machine learning applications and provide policymakers and public health experts with practical insights. This work establishes proactive air quality management techniques and emphasizes the importance of machine learning in solving air pollution's complex difficulties by determining the best models.

Individual contribution and findings: In the project, my primary responsibility involved normalizing and scaling the dataset. I employed various imputers, such as Iterative Imputer and KNN Imputer, and through experimentation, determined that KNN Imputer yielded the most optimal results.
Moving on to feature scaling, I implemented a range of techniques, including Min-Max Scaling, Standardization, Robust Scaling, Log Transformation, Power Transformation, and Decimal Scaling. Following a thorough comparison of Mean Squared Error (MSE) scores for each technique, it was discerned that Power Transformer emerged as the most effective choice for our dataset. Consequently, it was applied to both the training and test sets, ensuring the enhancement of overall model performance.

Individual contribution to project report preparation: In Section 2, Methodology and Proposal, I systematically presented the design model, employing a comprehensive flow chart for clarity. Additionally, I delved into the intricacies of the implementation platform and highlighted the key libraries utilized in the project. The data set was thoroughly described, accompanied by a comprehensive overview of the Indian Air Quality Index (AQI) Ranges.
Furthermore, I elucidated the various normalization techniques incorporated into the project, providing insights into their application. Subsequently, I explored the intricacies of Feature Scaling techniques and their impact on our data, shedding light on their role in enhancing the overall project outcomes."

Individual contribution for project presentation and demonstration: In the presentation, I demonstrated the need for normalization and feature scaling



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INDIVIDUAL CONTRIBUTION REPORT:

Air Quality Index Prediction Using Machine Learning Algorithm

ADARSH KUMAR
2029196

Abstract: Accurate Air Quality Index (AQI) prediction is essential for public health management as global air quality concerns grow. This work uses a collection of air quality metrics, meteorological factors, and pollutant concentrations to anticipate AQI values using machine learning regression models. Air pollution is rising due to factors including automotive emissions, industrial activities, and fossil fuel burning. This study examines the pros and cons of each regression model for AQI prediction through thorough experimentation and comparison. The findings expand environmental science machine learning applications and provide policymakers and public health experts with practical insights. This work establishes proactive air quality management techniques and emphasizes the importance of machine learning in solving air pollution's complex difficulties by determining the best models.

Individual contribution and findings: My main responsibility in the project revolved on doing site scraping and building the AQI prediction model. I performed a pivotal role in collecting crucial data through web scraping, guaranteeing that the model had a strong and solid basis. In addition, I explored the complexities of implementing models, utilizing a range of machine learning approaches. By doing thorough testing, I concentrated on adjusting hyperparameters to enhance the model's performance. The results of these endeavors made a substantial contribution to the overall success of the project by delivering a dependable tool for predicting the Air Quality Index (AQI).

Individual contribution to project report preparation: In my work, I directed my attention towards clarifying the complexity associated with AQI prediction result analysis, model implementation, and hyperparameter optimization. My efforts involved delivering thorough graphs, fine-tuning hyperparameters, and evaluating the gathered data. In addition, I enhanced the visual elements of the report by including key visualizations that correlate to the project's methodologies and findings.

Individual contribution for project presentation and demonstration: During the presentation, I played a significant role in its preparation. Specifically, I referenced several sources of information and contributed to the results section of the presentation. I seamlessly integrated key findings, fostering a clear understanding of the project's methodology and results using plots.



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INDIVIDUAL CONTRIBUTION REPORT:

Air Quality Index Prediction Using Machine Learning Algorithm

SHIVENDRA SINGH
2029197

Abstract: Accurate Air Quality Index (AQI) prediction is essential for public health management as global air quality concerns grow. This work uses a collection of air quality metrics, meteorological factors, and pollutant concentrations to anticipate AQI values using machine learning regression models. Air pollution is rising due to factors including automotive emissions, industrial activities, and fossil fuel burning. This study examines the pros and cons of each regression model for AQI prediction through thorough experimentation and comparison. The findings expand environmental science machine learning applications and provide policymakers and public health experts with practical insights. This work establishes proactive air quality management techniques and emphasizes the importance of machine learning in solving air pollution's complex difficulties by determining the best models.

Individual contribution and findings: My work in the project was to analyze different models and implement them to find the best suitable one for our project. I worked tirelessly on them to finalize the methods and methodologies that will increase the results significantly.

Individual contribution to project report preparation: In the report section I wrote the introduction,conclusion and the future scope by studying in detail regarding the same and analyzing different possibilities.

Individual contribution for project presentation and demonstration:
I explained the current challenges and constraints of the project and how we overcame them and made our project effective also explained in detail what current practices were lacking and what we did better to overcome them.



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